

Minimal pairs and Sound distribution

Unit 3, Class 2

NAMS

2025-07-27



Minimal pairs

- Two words that are identical except for one sound each.
- Replacing one sound, a word in the minimal pair can be changed into another.
- If two words are minimal pairs differing only in [X] and [Y], then they are said to be minimal pairs for [X] and [Y].
- If there are minimal pairs of [X] and [Y], then they are allophones of different phonemes.



Minimal pairs: Examples

- Minimal pairs of [k^h] and [b] in English test:
 - ‘cat’: [k^hæt], ‘bat’: [bæt]
- Minimal pairs of [p] and [p^h] in Nepali:
 - ‘parsi’: [pʌrsi], ‘pharsi’: [p^hʌrsi]
- Not minimal pairs of [p] and [p^h] in English:
 - ‘pan’: [p^hæn], ‘span’: [sp^hæn]
- In Nepali, [p] and [p^h] are allophones of different phonemes /p/ and /p^h/, but in English they are allophones of the same phoneme /p/.



- Set of positions sounds can occupy in an utterance
- A part of knowledge of language is the knowledge of restrictions on distribution



- If two phones [X] and [Y] distinguish two words in a minimal pair, they are in contrastive distribution.
- [p] and [p^h] are in contrastive distribution in Nepali ([pʌrsi] vs [p^hʌrsi]) but not in English ([p^hæɪn] and [sp^hæɪn]).



Complementary distribution

- If there are no minimal pairs for the phones [X] and [Y], they are in complementary distribution.
 - Native speakers follow the implicit distribution rule where one allophone or the other occurs.
 - e.g., [p] and [p^h] in English
 - /p/ is aspirated when it is the first sound in a syllable (onset).
- Allophones of the same phoneme are in complementary distribution.
 - e.g., allophones of /d/ in Nepali
 - 'mād': [ma:ɽ], 'damaru': [dʌmrɯ] → /d/
- Replacing one for the other won't necessarily change the meaning of the word.
 - But one doesn't occur in the environment/context where other occurs.



- Phenomenon of two (or more) sounds occurring in the same environment without changing the meaning of the word
 - not considered incorrect by native speakers of the language
- route: [ru:t] and [raʊt] in English
- ain: [ʌin] vs [en] in Nepali*

*[en] is considered incorrect by many speakers of Nepali.

